Correlation and Regression

Regression

Regression: The statistical tool with the help of which we are in a position to estimate (predict) the unknown values of one variable from known values of another variable is called regression.

Linear Regression: If the variable X and Y in a bivariate distribution are related. We will find that the points in a scatter diagram will cluster around a curve called **regression curve.** If the curve is a **straight line**, it is called the line of regression and the regression is known as **linear regression**.

The two Variable Regression Model

Let $(x_1, y_1), (x_2, y_2), \dots\dots\dots\dots. (x_n, y_n)$ be n pair of observations from a vibarite population. Then the regression model of the two variables can be expressed as:

$y_i = \alpha + \beta x_i + \epsilon_i; i = 1,2,3, \dots\dots. n.$

Where,

y=Dependent variable

x=Independent variable

α *and* β are parameters and ε is random error.

Assumptions:

i. $\varepsilon's$ are normally distributed
ii. $E(\varepsilon_i) = 0$
iii. $E(\varepsilon_i, \varepsilon_j) = 0 ; i \neq j$
iv. $E(\varepsilon_i{}^2) = \sigma^2$
v. *The values of the independent variable must be fixed.*

Regression Analysis: Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of the original units of the data.

The main purpose of the regression analysis: The main purpose of simple regression analysis is to:

i. Establish a functional or mathematical relationship between dependent (response) and independent (predictor) variables.
ii. Estimate or predict the value of the dependent (response) variable for given value of the independent (predictor) variable
iii. Show the pattern or trend of the dependent variable for various value of the independent variable.

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Correlation and Regression

Important properties of regression coefficient:

i. The coefficient of correlation is the geometric mean of the two regression coefficient
ii. If one of the regression coefficients is greater than one, then the other must be less than one since $r_{xy}^2 = b_{x/y} . b_{y/x} \leq 1$.
iii. Both the regression coefficient will have the same sign
iv. The sign of the correlation coefficient is same as the regression coefficients
v. The average value of the two regression coefficients would be greater than the value of coefficient of correlation
vi. Regression coefficients are independent of change of origin but not scale
vii. Regression coefficient can take any real value between $-\infty \; to + \infty$

Multiple Regression Model: Most application of regression analysis involve the use of more than one explanatory variable. Regression models that employ more than one independent variable are called multiple regression model. When all parameters enter the equation linearly it is called multiple linear regression model.

We write the multiple regression model as

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \epsilon_i ; i = 1,2,3, \ldots \ldots . n.$$

Where Y is the dependent variable, the X's are the independent variables and ε is the error term. Here we have a dependent variable and k explanatory variables excluding the intercept term. This model is also called a k+1 variable regreesion model.

Regression assumptions:

The assumptions of the multiple regression model are quite similar to those of the two variable linear regression model:

i. The relationship between Y and X is linear. But no exact linear relationship exists between two or more X's.
ii. The X's are non stochastic variables are fixed.
iii. The error term has expected values: $E(\varepsilon) = 0$
vi. The error term has constant variance for all observations, i.e., $E({\varepsilon_i}^2) = \sigma^2$ $; i = 1,2,3, \ldots \ldots . n.$
vii. The random variables ε_i are statistically independent. Thus $E(\varepsilon_i, \varepsilon_j) = 0 \, ; for \; all \; i \neq j$
iv. The error term is normally distributed.

Polynomial Regression: Polynomial models are useful in situations where the analyst knows that curvilinear effects are present in the response function. In general the k-th order polynomial model in one variable is

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \varepsilon.$$

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Correlation and Regression

Correlation

Correlation: The statistical tool with the help of which these relationships between two or more than two variables can be studied is called correlation.

A M Tuttle gives a very simple definition of correlation. According to A M Tuttle, an analysis of co variation of two or more variables is usually called correlation.

Note: The linear relationship between two variables is defined as correlation.

According to the nature of relation between two variables, correlation can be divided as

i. Perfect positive correlation
ii. Partial positive correlation
iii. Zero correlation
iv. Perfect negative correlation and
v. Partial negative correlation.

Correlation coefficient: A measure of intensity or degree of linear relationship between two variables is called coefficient of correlation.

Let $(x_i, y_i); i = 1,2,3, \dots \dots, n$ be a set of n pair of observations from a bivariate population of the random variables x and y. Then the Karl Pearson's correlation coefficient is denoted by r_{xy} or simply r and defined as

$$r_{xy} = r = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\Sigma(x_i-\bar{x})^2\,\Sigma(y_i-\bar{y})^2}} = \frac{\Sigma x_i y_i - \frac{\Sigma x_i \Sigma y_i}{n}}{\sqrt{\left\{\Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}\right\}\left\{\Sigma y_i^2 - \frac{(\Sigma y_i)^2}{n}\right\}}}.$$

[see page - 315 & 323]

Important properties of correlation coefficient:

i. The correlation coefficient r lies between -1 and +1
ii. It measures the magnitude and direction of statistical relationship between two variables
iii. It is a pure number. That is, it is independent of the units of the measurement of the variables
iv. It is symmetrical function of x and y, that is $r_{xy} = r_{yx}$
v. $r = 0$ indicates no linear relationship between x and y
vi. $r = +1$ indicates a perfect positive relationship between x and y
vii. $r = -1$ indicates a perfect negative relationship between x and y
viii. The geometric mean of the two regression coefficients is equal to the correlation coefficient
ix. The correlation coefficient is independent of the shift of origin and change of scale.

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The main purpose of the Correlation: The main purpose of Correlation is to:

i. Correlation is used to describe the linear relationship between two continuou variables (e.g., height and weight). In general, correlation tends to be used whe there is no identified response variable. It measures the strength (qualitatively) an direction of the linear relationship between two or more variables.

ii. The Pearson correlation coefficient measures the strength of the linear associatior between two variables. It can be used to estimate the population correlation, ρ.

Difference between correlation coefficient and regression coefficient.

Regression	Correlation
Regression means determination of estimated value of the independent variable with respect to one or more independent variables.	Correlation means the linear relationship of two or more dependent variables with each other.
In regression influenced variable is called dependent variable and other variables are independent.	In correlation the variables are dependent on each other variable. There is no concept of independent variable.
In regression, only the dependent variable needs to be random.	In correlation, both the variables must be random.
In regression, either $y = f(x) or\ x = f(y)$ or both may be true.	In regression, either $y = f(x)\ and\ x = f(y)$ are to be true.
$b_{xy} \neq b_{yx}$	$r_{xy} = r_{yx}$
It has no upper and lower limits	$-1 \leq r \leq +1$
Regression coefficients are independent of change of origin but not of scale.	It is independent of both change of origin and scale.

Example of regression: Page 330-331, book Business statistics, MKR.

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